

Esha Manchanda

esha006@e.ntu.edu.sg | Academic Webpage

EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY

PHD (LEE KONG CHIAN SCHOOL OF MEDICINE)

Data Science & Artificial Intelligence Research Programme

Topic: Deep learning and computational modelling approaches to investigate cortical pathology in a rat model of Multiple Sclerosis using spatial transcriptomics.

Singapore

Aug 2025 - Present

NANYANG TECHNOLOGICAL UNIVERSITY

MASTERS IN BIOMEDICAL DATA SCIENCE

Artificial Intelligence Track

GPA: 4.92/5.0

Singapore

Aug 2024 - May 2025

ASHOKA UNIVERSITY

POSTGRADUATE DIPLOMA IN ADVANCED STUDIES AND RESEARCH IN COMPUTER SCIENCE

GPA: 3.83/4.0

Graduated Magna Cum Laude with Batch Rank 1

India

Aug 2023 - May 2024

ASHOKA UNIVERSITY

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

GPA: 3.77/4.0

Graduated Magna Cum Laude

India

Aug 2020 - May 2023

RESEARCH EXPERIENCE

LKC MEDICINE NANYANG TECHNOLOGICAL UNIVERSITY | PHD CANDIDATE

August 2025 – Present | Singapore

- Glioblastoma Tumour Heterogeneity Study: Analysing single-cell and spatial transcriptomic data to characterise malignant cell state architecture in Glioblastoma.
- Angptl4 Neutrophil State Study: Built a human neutrophil atlas from public single-cell RNA-seq datasets. Conducted ligand–receptor communication analysis to map neutrophil–macrophage signaling networks.

LKC MEDICINE NANYANG TECHNOLOGICAL UNIVERSITY | RESEARCH INTERN

June 2024 – July 2025 | Singapore

- Investigated the cross-talk between metabolic dysfunction and inflammation in the pathophysiology of progressive Multiple Sclerosis (MS) at [Lo Lab](#).
- Analyzed bulk RNA-seq data to identify metabolic genes that were dysregulated in MS.
- Analyzed single-cell RNA sequencing data to determine how metabolic gene expression varies across cell types in the MS brain.

ASHOKA UNIVERSITY | UNDERGRADUATE THESIS CANDIDATE

Sept 2023 – May 2024 | India

- Developed a mathematical model to explore the role of T-cell exhaustion in Epstein-Barr virus-mediated dysregulation of B cells and its potential contribution to chronic inflammation in MS.
- [Thesis Manuscript](#)

INTEGRATIVE GENETICS LAB ASHOKA UNIVERSITY | RESEARCH INTERN

August 2023 – March 2024 | India

- Conducted differential gene expression analysis on RNA sequencing data to investigate dietary impact on gene expression in fruit flies.
- Identified specific genes that exhibited upregulation or downregulation in response to changing protein:carbohydrate ratios, and determined the differential activity of biological pathways based on the diet.

CENTRE FOR BIOINFORMATICS & COMPUTATIONAL BIOLOGY ASHOKA UNIVERSITY |

RESEARCH INTERN

Jan 2023 – June 2023 | India

- Implemented the Model-Agnostic Meta-Learning (MAML) framework coupled with few-shot learning for disease classification from human gut microbiome data.
- Conducted comparative analysis of MAML against traditional neural networks, demonstrating MAML's superior adaptability in handling high-dimensional, heterogeneous data.
- [Project Report](#)

COMPUTATIONAL HEALTH SCIENCES LAB ASHOKA UNIVERSITY | RESEARCH INTERN

Aug 2022 – Dec 2022 | India

- Conducted a review of algorithms for Single-cell Assay for Transposase-Accessible Chromatin sequencing (scATAC-seq) analysis to assess the computational challenges associated with scATAC-seq analysis including inherently sparse data, determination of features, peak calling, impact of sequencing coverage and noise, and clustering performance.

PUBLICATIONS

Data-Efficiency with Comparable Accuracy: Personalized LSTM Neural Network Training for Blood Glucose Prediction in Type 1 Diabetes Management

E. Manchanda, J. Zeng, C.H. Lo

[\[Paper\]](#)

Angptl4 Buffers Proteostatic Stress Thresholds to Control Neutrophil State Transitions.

(Co-author)

[\[In preparation\]](#)

POSTERS & TALKS

Singapore Symposium on Brain-Body Interactions 2024 & Neuroscience Singapore FEBS-IUBMB-ENABLE 2024

[\[Talk & Poster\]](#)
[\[Poster\]](#)

HONORS & AWARDS

Best Flash Talk , Singapore Symposium on Brain-Body Interactions & Neuroscience Singapore	2024
Best Capstone Thesis Award , Department of Computer Science, Ashoka University	2024
Gold Medallist , Department of Computer Science, Ashoka University	2024
Full Tuition Scholarship , Bachelor of Engineering in Computer Science, University of Hong Kong	2020

TEACHING

Deep Learning for Biomedical Sciences , Teaching Assistant	Sem 1 AY 2025-26
Computational/Mathematical Biology , Teaching Assistant	Spring 2024
Introduction to Computer Programming , Teaching Assistant	Spring 2022
Computational/Mathematical Biology , Teaching Assistant	Fall 2021
IEEE Ashoka Computational Biology PySci , Workshop Instructor	Spring 2024
Computer Science Society Tools for Web Scraping , Workshop Instructor	Fall 2021

TECHNICAL SKILLS

Programming	Python, R, C, Markdown, $\LaTeX$
Frameworks	PyTorch, Tensorflow, Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, ggplot
Bioinformatics	Scanpy, Seurat, Cytoscape, Tangram, GraphST, DESeq2

LANGUAGES

Proficient	English, Hindi
Intermediate	Korean